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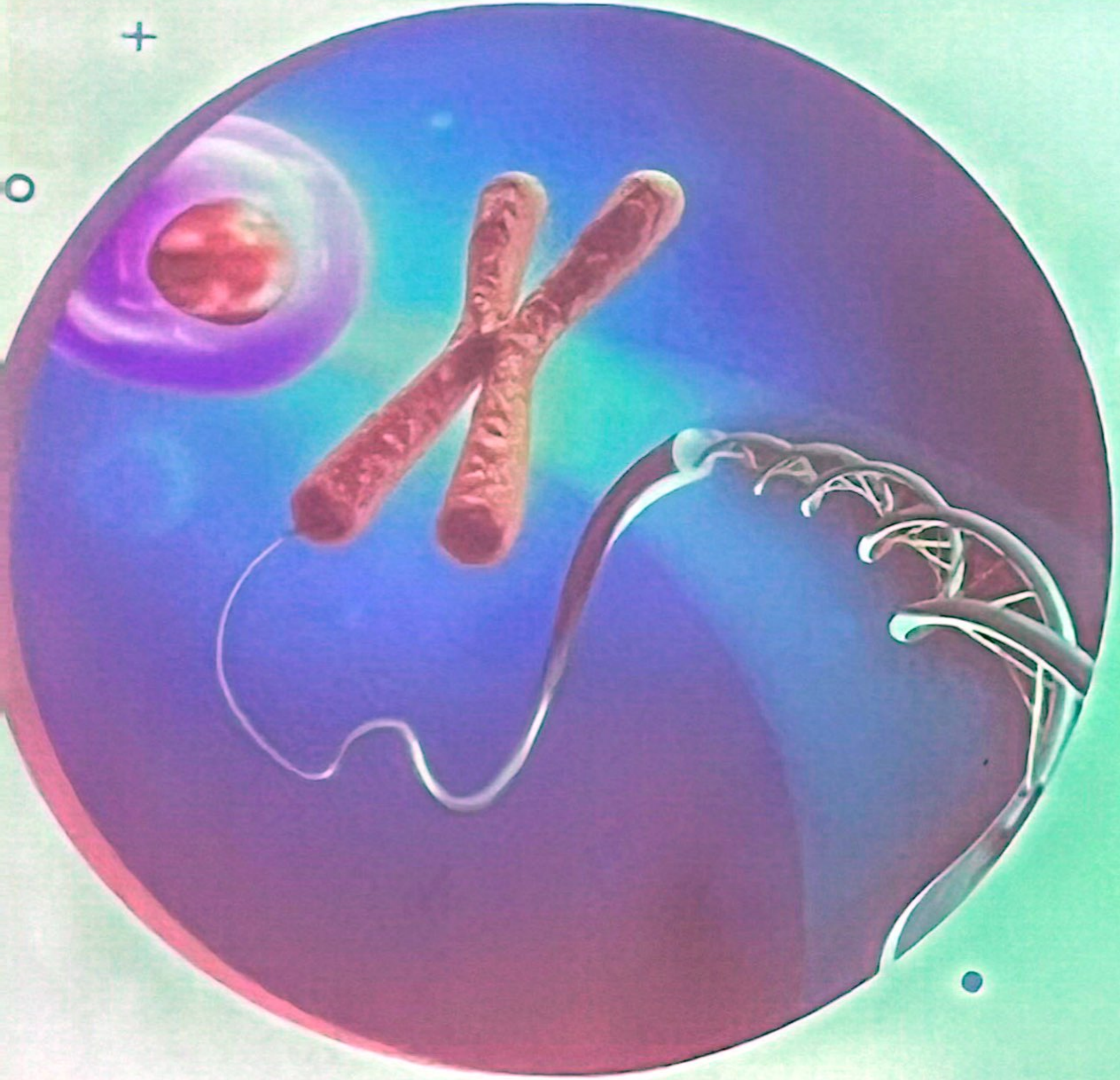

Lectuer 11

Inherited Diseases of biochemical origin

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Outlines

- **Inherited Diseases of biochemical origin**
  - I. Inborn Errors of Metabolism
    1. Inherited Errors in metabolism of phenylalanine
    2. Congenital Hemolytic Anaemia due to Pyruvate Kinase deficiency
    3. Inherited lysosomal Storage Disease
    4. Porphyria (Pink Tooth)
    5. Dermatosparaxis
  - II. Inherited Bleeding Disorders
  - III. Inherited Haemoglobin Disorders
- **Genetic Resistance and Pathogen**
- **Genetic Control of Inherited Diseases**

Inherited Diseases of biochemical origin

All simply inherited disorders and diseases → have a biochemical origin

↑

Gene Mutation

↓ ↓

Simplest alteration in polypeptide structure

Complete failure of cell to synthesize the polypeptide

↓

Mutant polypeptide

↓

Unable to do its specific function properly

↓

The *physiological process* which required polypeptide become impaired

Inherited Diseases of biochemical origin

I. Inborn Errors of Metabolism

II. Inherited Bleeding Disorders

III. Inherited Haemoglobin Disorders

I. Inborn Errors of Metabolism

Mutation in Gene

Affect polypeptide which act as enzyme or part of enzyme

↓

Deficiency of that Enzyme

↓

A blockage in metabolic pathway

↓

Faulty metabolism and occumelation of harmfull metabolites

↓

$\boxed{\text{Diseases}}$

Inborn Errors of Metabolism

I. Inborn Errors of Metabolism

A~~A~~ $\xrightarrow{\text{mutation}}$ Deficiency in the enzyme will not show in heterozygote

Give enzyme in normal level which do normal function

↓

Because normal allele will produce enough enzyme

So , no defect may be observed on heterozygous individual except the amount of enzyme less than in normal homozygous individual

We can detect **heterozygous carriers** by measuring the enzyme activity in blood or any cell (heterozygous detection) 50%

I. Inborn Errors of Metabolism

1. Inherited Errors in metabolism of phenylalanine

2. Congenital Hemolytic Anaemia due to Pyruvate Kinase deficiency

3. Inherited lysosomal Storage Disease
   a- Tay-Sack´s Disease (Human)
   b- Mannosidosis (Cattle)

4. Porphyria (Pink Tooth)

5. Dermatosparaxis

I. Inborn Errors of Metabolism

1. Inherited Errors in metabolism of phenylalanine

Six different inborn diseases $\xrightarrow{\text{caused by}}$ Seven autosomal recessive mutation

↓

blockage in metabolic pathway of phenylalanine

Normal pathway	Disorders
Phenylalanine → Tyrosine	Phenylketonuria Mental disorder
Tyrosine → P-hydroxyphenyl pyruvic acid	Tyrosinosis
P-hydroxyphenyl pyruvic acid → 2,5 dihydoxy phenyl pyruvic acid	Tyrosinemia
2,5 dihydoxy phenyl pyruvic acid → Homogentisic acid	Alkaptonuria
Tyrosine → Thyroxine	Cretinism
Tyrosine → DOPA	Albinism
DOPA → Melanin	Albinism

I. Inborn Errors of Metabolism

2. Congenital Hemolytic Anaemia due to pyruvate kinase deficiency

- Dog (Basengi and Beagle breed)

- Cause → Deficiency of pyruvate kinase enzyme which need for anarobic glycolysis (glycogen to glucose)

Lead to
- decreased glucose utilization
- decreased level of adenosine triphosphate (ATP)

- Weakness
- Excessive sleeping

Ex. of poleiotropy

↓

Decreased life span of red blood cells (Congenital Hemolytic Anaemia)

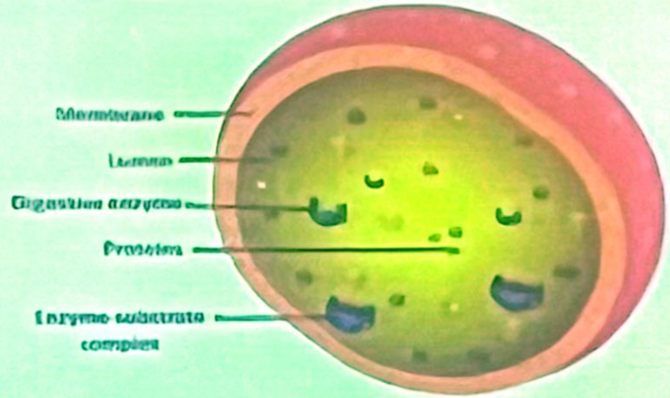
➢ Heterozygous doge (Dd) normal external symptoms but level of PK is 50%

↓

Heterozygous detection

I. Inborn Errors of Metabolism

3. Inherited Lysosomal Storage Disease



Lysosome

- Small memmbran bound organelles present in cytoplasm

- Contain a great variety of enzymes which work in step by step to breakdowen (catabolism) of diffrent complex molecules (fat, carbohydrate, protien and nuclic acid) to simple form

? If enzyme absant or in active → Degredation is stopped → storage

➤ Inborn Errors of lysosomal catabolism → Lysosomal Storage Disease

3. Inherited Lysosomal Storage Disease

Ex. Mannosidosis (Angous Cattle)

Caused by autosomal recessive gene

↓

Recessive homozygote lack of enzyme *α-mannosidase*

(required for metabolise saccharides containing mannose)

↓

Mannose acccumulate and cause lysosomal storage disease

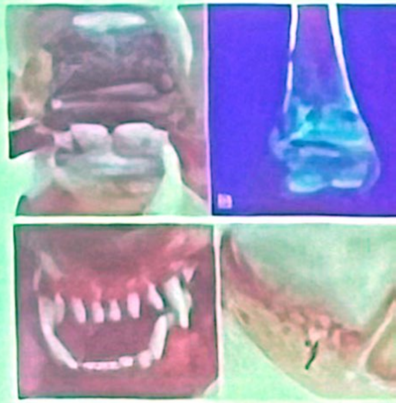
↓

Death within first years

I. Inborn Errors of Metabolism

4. Porphyria (Pink Tooth)

- Reported in human, cattle, cat and pig



- Error in metabolism associated with defficient in synthesis of heme
- Caused by recessive mutation in autosomal gene

(*uroperphyrinogen III cosynthetase*)

Lack of heme and accumulate in the body intermediates of heme metabolism

↓ Lead to hemolytic anemia

↓ *uroperphyrinogen I*

Lead to red staining of tooth, bone, urine

Prophyrinogens → oxidized to prophyrin

↓

Absorb visibal light and increase photosensitivity

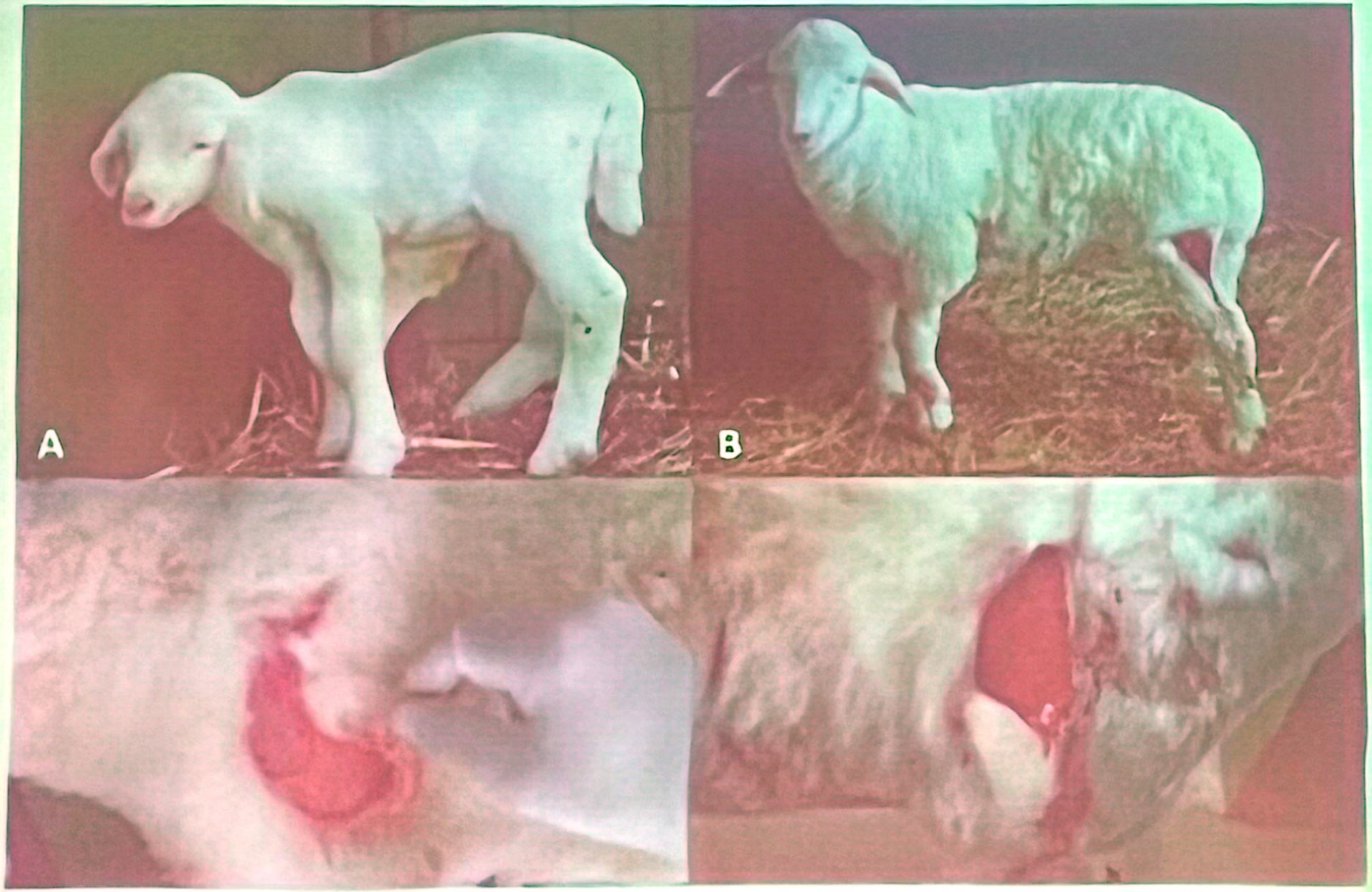
- Heterozygous carrier of a mutant gene have enzyme activity intermediate between normal homozygotes and diseased individual

I. Inborn Errors of Metabolism

5. Dermatosparaxis

- Disease in sheep and cattle
- Caused by recessive mutation in autosomal gene (*Procollagen peptidase*)
- Heritable disorder of connective tissue due to error in metabolism
- Homozygous recessive individuals → formed abnormal procollagen (flattened twisted ribbons) make skin very week and easily torn
- Symptomes → - Skin easily extendible and very fragil
  - Sever laceration with lightest scratch

5. Dermatosparaxis



5. Dermatosparaxis

...s et al., 2002

II. Inherited Bleeding Disorders

- Defficiency in one of **coagulation factor** →

  Interfere with normal coagulation of blood or onset of bleeding

- **Hemophilia A** → - Cat, Horse and Human
  - Caused by defficiency of coagulation factors VIII
  - Its due to an ***X-linked recessive gene***

- **Hemophilia B** → - Cat and Dog
  - Caused by defficiency of coagulation factors IX
  - Its due to an ***X-linked recessive gene***

II. Inherited Bleeding Disorders

- **Inherited Bleeding Disorders other than hemophilia**

  - Due to an ***autosomal codominant or incomplete dominant***

  - Most of this condation were described in Dog and Cat

  - Due to defficiency of coagulation factors I, II, VII,VIII, X, XI, and XII

↓

(Psedudohemophilia)

These diseases and partial deficiency of the factors can detected biochemically

III. Inherited Haemoglobin Disorders

- Mutation in genes specifying haemoglobine chains

– Identified in hunman and animals

↓

Defective haemoglobine idintified as cause of haemoglobine disorder)

↓

Most femous example is sickle cell anemia

↓

Inherited in codominant pattern


Genetic
Resistance
and
pathogens

Genetic Resistance and pathogens

Pathogens

Bacteria Parasites Virus

↓

Influence on animal production

Genetic Resistance and Pathogens

Studies have been made in genetic control of:

1- Genetic Resistance in Hosts

a. Resistance to Marek's Disease in Chickens
b. Resistance to Worms in Sheep
c. Resistance to Ticks in Cattle

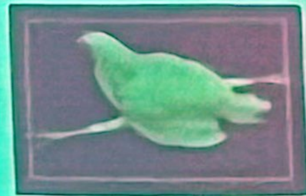
2- Genetic Resistance in pathogen

a. Resistance to Insecticides in Insect Parasites
b. Resistance to Anthelmintics in Worms
c. Resistance to Antibiotics in Bacteria

1- Genetic resistance in host

➢There are many examples of genetic variation in hosts for resistance to pathogens:

a. Resistance to Marek's Disease in Chickens



Disease association with MHC

- is a set of cell surface molecules encoded by a large gene family which controls a major part of the immune system in all vertebrate
- The major function of MHCs is to bind to antigenic portion of pathogens and display them on the cell surface for recognition by the appropriate T-cells

➢ There is strong association between B^{21} allele (MHC locus) and resistance

1- Genetic resistance in host

b. Resistance to Worms in Sheep

- There is evidence of variation between breeds and genetic variation within breed in resistance to worm infestation in sheep

- Genetic differences within breeds are shown by estimating heritability which open the way for selection programs aimed to increasing resistance

> Is a statistic used in the fields of breeding and genetics that estimates the degree of *variation* in a *phenotypic trait* in a population that is due to genetic variation between individuals in that population

- In an experiment in Australia, the heritability of 0.35 was estimated in Merino sheep for maximum worm eggs/gm of feces.

2- Genetic resistance of host against the pathogen

c. Resistance to Ticks in Cattle

Bos indicus (Indian) cattle are much more resistant to ticks than Bos taurus (European) cattle raised under the same conditions

Also, crosses between the two species are more resistant to ticks than European cattle

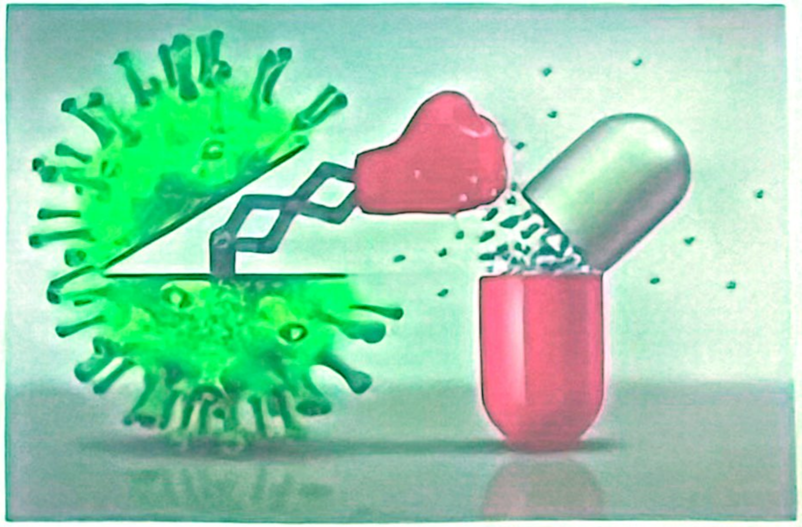
The heritability of resistance to ticks is very high (80%) among the crosses between the two species, which indicates that selection would be very effective

3- Genetic resistance in pathogen

a. Resistance to Insecticides in Insect Parasites

b. Resistance to Anthelmintics in Worms

c. Resistance to Antibiotics in Bacteria



a. Resistance to Insecticides in Insect Parasites

The wide application of chemical insecticides as:

- DDT
- Organophosphorus compounds

have after led to fast preliminary results followed later by *partial* or *complete ineffectiveness* of the insecticide

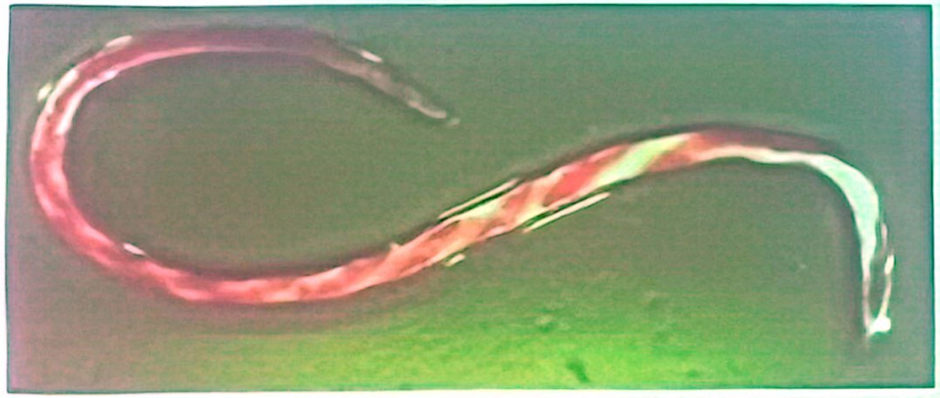
Reason?

→ The same results were obtained with the use of chemical against ticks in cattle (acaracides)

b. Resistance to Anthelmintics in Worms

The same principle apply

For example, the use of **anthelmintic thiabendazole**, which is a very **strong** anthelmintic against round worms, has led after only three years to the development of **resistant strains** of the very harmful worm *Haemonchus controtus*



c. Resistance to Antibiotics in Bacteria

- Widespread use of different antibiotics against bacteria led to appearance of resistant strains

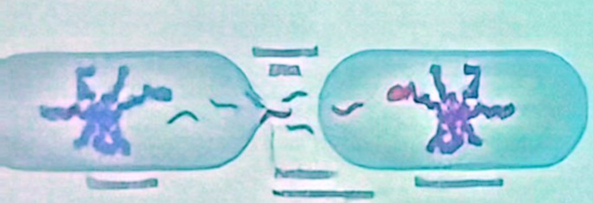
- Many of these strains are resistant to more than one antibiotic

In addition to previous mention method in resistance of parasite, antibacterial resistance to Antibiotic **very rapid** due to ability of bacteria to transfer resistant genes horizontally **(between contemporary individuals, within the same generation)** as well as vertically **(from one generation to the next).**

There are three methods by which bacteria can transfer genes horizontally:

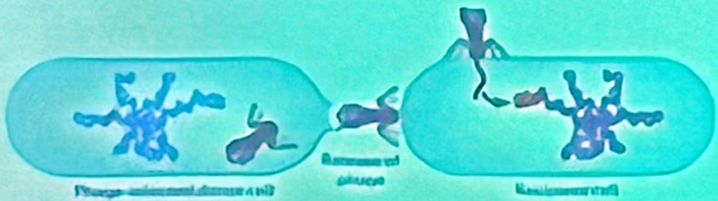
1. Transformation

The release of naked DNA from one cell, its uptake by another cell



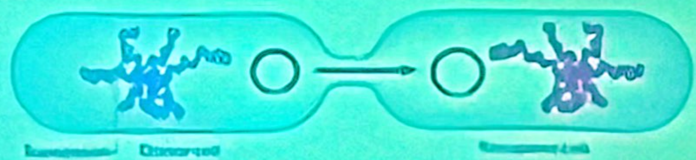
2. Transduction

The transfer of DNA from cell to cell by means of bacteriophage



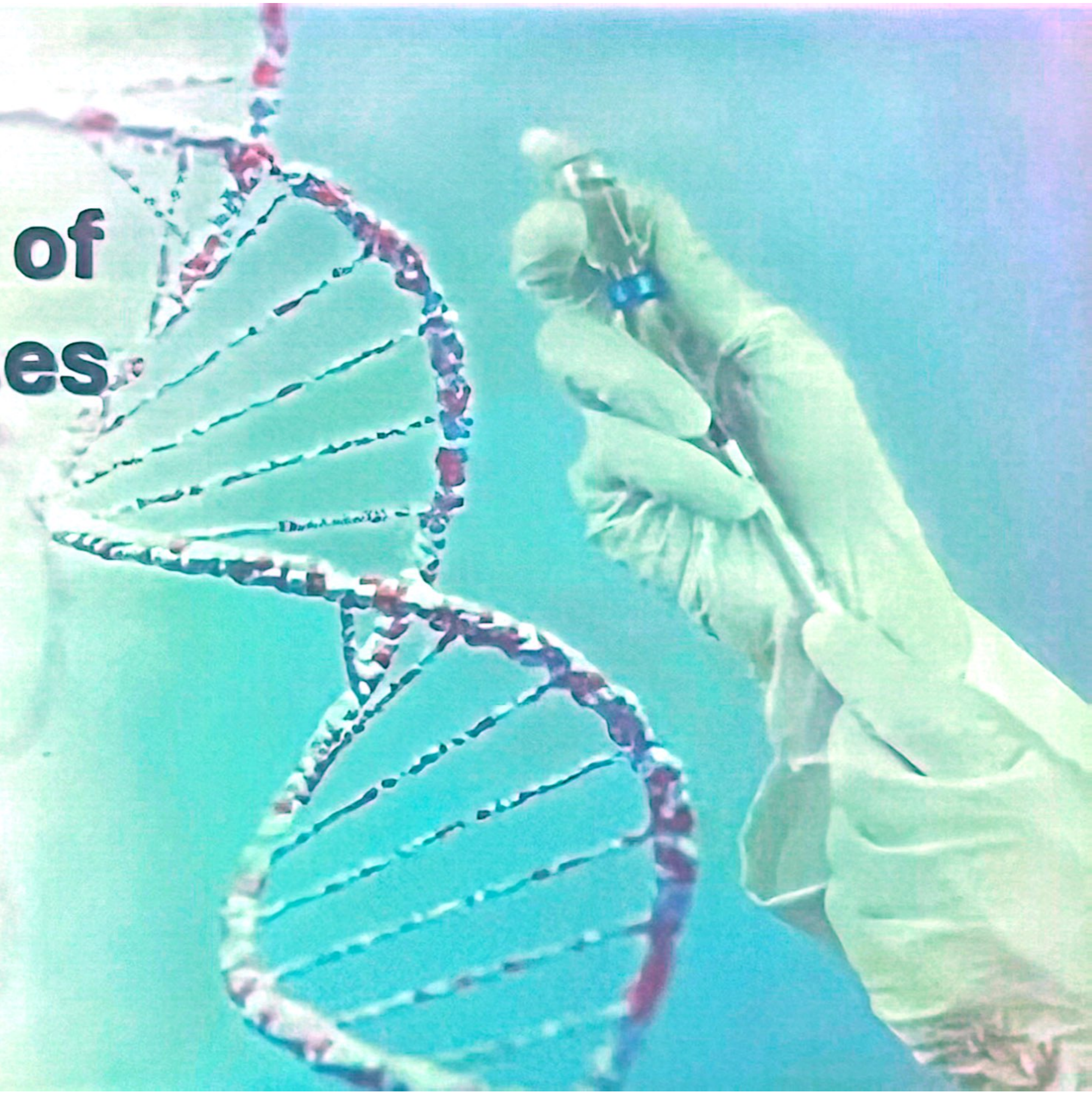
√ **3. Conjunction**

- The transfer of DNA from one cell to another following the joining together or mating of the two cells



- Many important genes for antibiotic resistance are located on plasmids. Those plasmids that carry one or more resistance genes are called **R. factors**.

Genetic Control of Inherited Diseases



2- Genetic Control of Inherited Diseases

1- Detection the individual which have inherited diseases

2- Treatment of inherited diseases

a. Gene therapy

b. Enzyme replacement therapy

1- Detection the individual which have inherited diseases

- Inherited disease controlled by single gene locus → easier to control

➢ Affected individual or heterozygous → Culled

Detection of heterozygotes

'edigree or test cross

n case of recessive disease

Biochemical screening

In case of inborn error of metabolism

(Mutation in gene specifying polypeptide Which act as enzyme or part of enzyme)

DNA screening

depend on presence of DNA polymorphism

Cause by difference in Nucleotide sequence between mutant and normal one

a. Gene therapy

An experimental technique for correcting defective genes that responsible for disease development

Types

- Somatic gene therapy
- Germ-line gene therapy

Somatic gene therapy

- Insertion of new genetic material (normal gene) into the genome of the same affected individual
- **Body cells** and **tissues** other than the reproductive ones

Steps:

1. Identify and isolate the normal gene at the relevant locus or construct it (if its a.a sequence is known)

2. Produce large no of copies of the gene by cloning

3. Insert foreign DNA into appropriate cells which previously removed from the patient

Somatic gene therapy

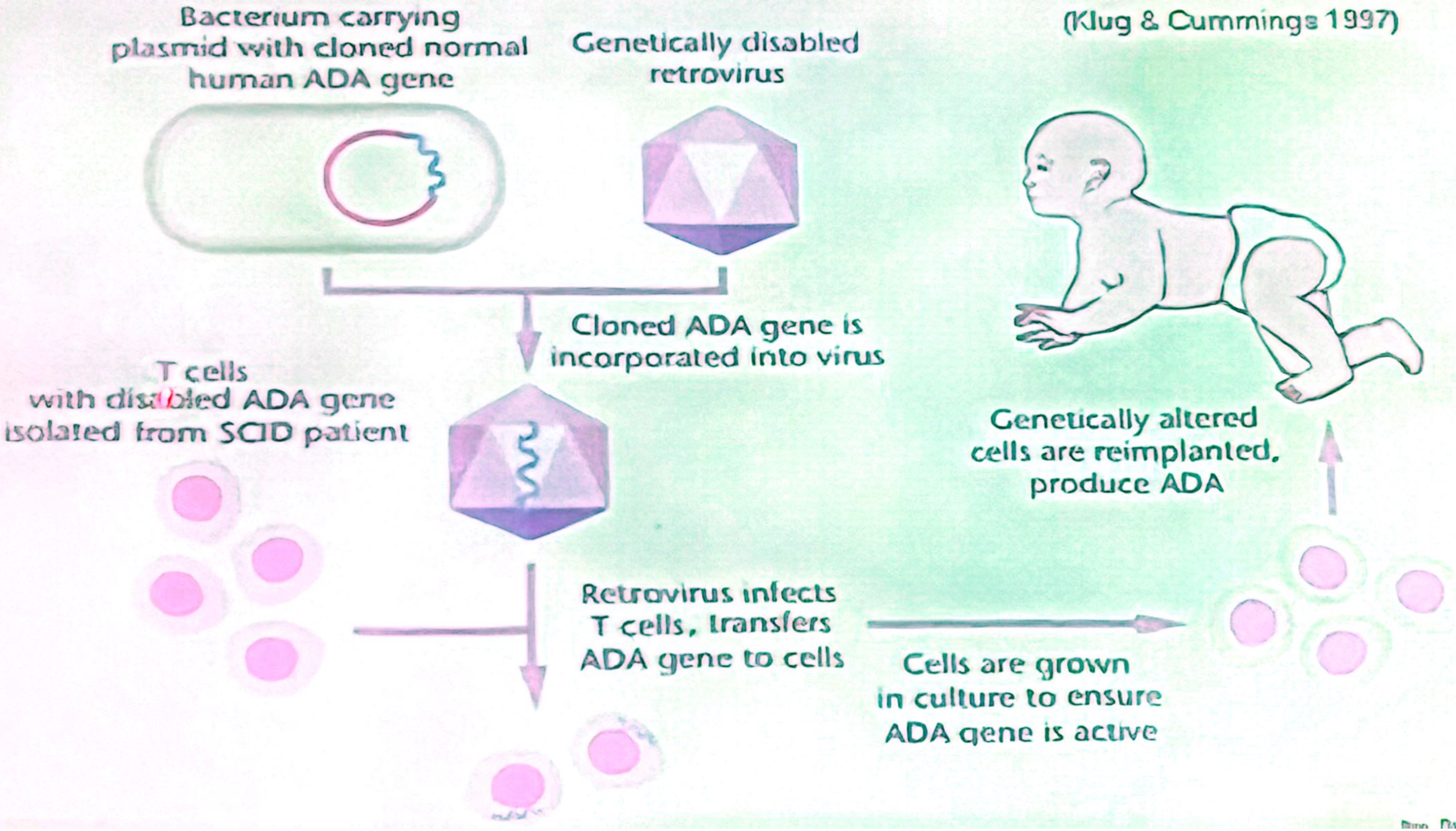
- There are **several method** can be used to transfer genes into individual cells
- This methods include used of virus as vectors, chemically assisted transfer of gene across cell membrane and fusion of cell with artificial vector contain cloned DNA

➢ Somatic gene therapy ⟶
- Apply with genes which expressed only in one tissue
- More important in human than in animals

(treating heritable genetic disorder)

Somatic gene therapy

Severe Combined Immunodeficiency (SCID)



Germ-line gene therapy

- It involves adding the relevant foreign DNA to:

a fertilized ovum ⟵ ⟶ an embryo

Comparasion

SOMATIC CELL GENE THERAPY	GERM LINE GENE THERAPY
Therapeutic genes transferred into the somatic cells.	Therapeutic genes transferred into the germ cells.
Eg. Introduction of genes into bone marrow cells, blood cells, skin cells etc.	Eg. Genes introduced into eggs and sperms.
Will not be inherited later generations.	It is heritable and passed on to later generations.
At present all researches directed to correct genetic defects in somatic cells.	For safety, ethical and technical reasons, it is not being attempted at present.

2- Treatment of inherited diseases

b. Enzyme Replacement Therapy

- supplying the affected individual with regular doses of the deficient enzyme or with normal cell or tissue transplants

One of the major disadvantages → the problem of foreign tissue rejection

- Gene therapy could overcome this problem, as the cells of the affected individual itself are removed and returned to *its* body after being "repaired".